

Simulation of Network Traffic under Negative exponential Interarrival time

Simulation of Network Traffic

The pdf of interarrival time of a negative exponential distribution is expressed like,

$$f(t) = \lambda e^{-\lambda t}; \text{ where } \lambda \text{ is call arrival rate and } t \text{ is inter-arrival time.} \quad (1)$$

Integrating (1) over $[0, t]$ gives the cdf,

$F(t) = rand()$; a random number in $[0, 1]$

$$t = -(1/\lambda) * \ln(1 - F(t)) \quad (2)$$

249 random numbers are generated; only 30 are shown in table-D.1

Here observation period, $T = 100$ units.

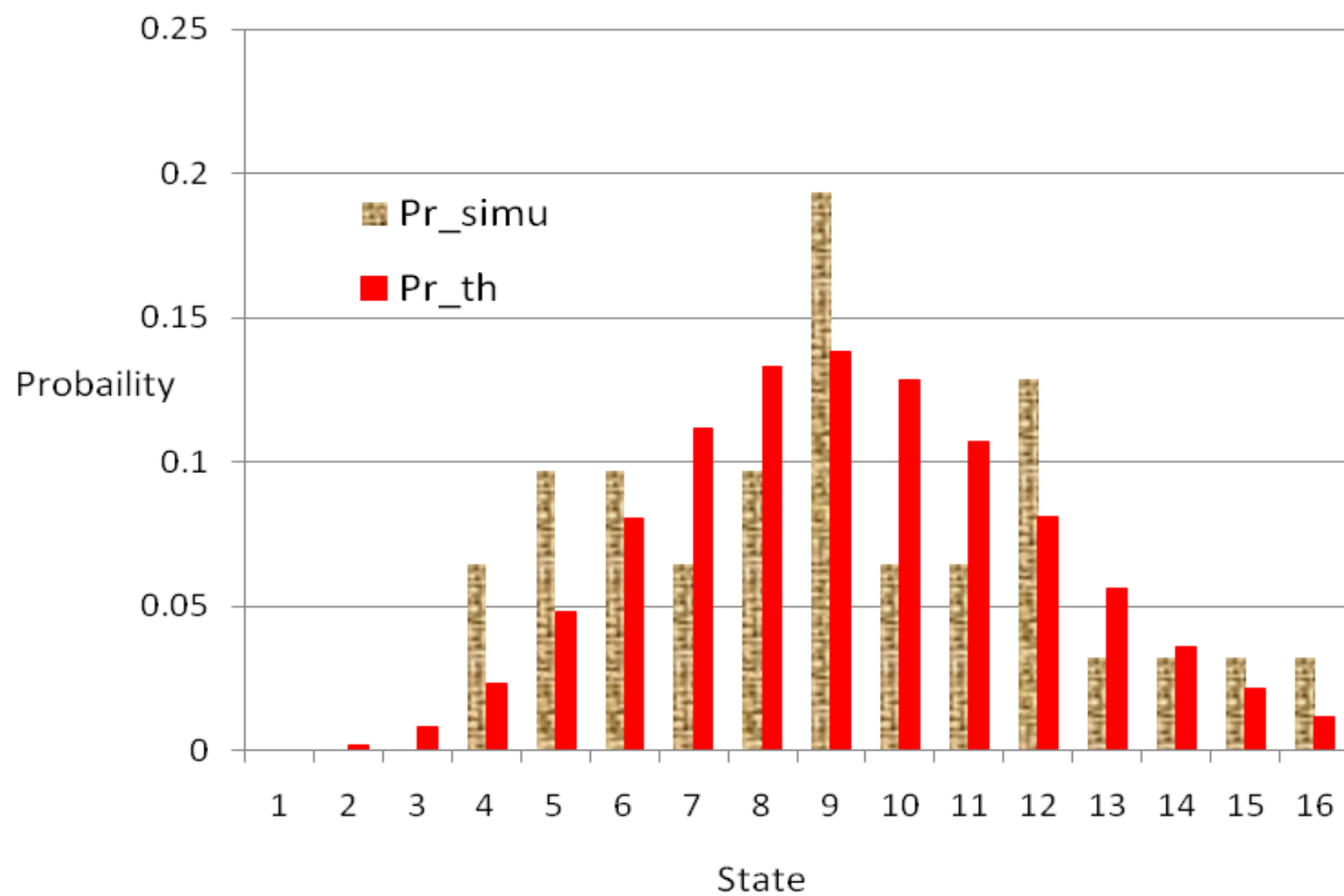
SL	CDF $F(t) = \text{rand}()$	Interarrival time $t = -(1/\lambda) * \ln(1 - F(t))$	Arrival Instant (t_i)
1	0.468521	7.585101703	7.585101703
2	0.007956	0.095852588	7.680954292
3	0.43571	6.866236391	14.54719068
4	0.095414	1.203333423	15.75052411
5	0.283396	3.998790785	19.74931489
6	0.430274	6.751192205	26.5005071
7	0.661874	13.01204213	39.51254923
8	0.258393	3.587236016	43.09978524
9	0.071614	0.891691955	43.9914772
10	0.055007	0.678937034	44.67041423
11	0.298864	4.260644398	48.93105863
12	0.329002	4.787861424	53.71892005

- ✓ Number of arrival in [0-100] minutes is 15, in [100, 200] is 3 and [200, 300] is 11 shown in table-D.2. The entire result is shown in table-D.2.
- ✓ Use excel formula, =COUNTIF(Range of arrival instant of table D1, "<" & cell of second column of table D2) to get the results of column 3 of table D.2. In above formula the 'Range of arrival instant of table D1' should be made fixed pressing F4. For example: COUNTIF(\$E\$4:\$E\$253,"<"&H5).

Time Range		Number of Calls till final time	Number of calls in the range
From	To		
0	100	15	15
100	200	18	3
200	300	29	11
300	400	40	11
400	500	45	5
500	600	54	9
600	700	62	8
700	800	66	4
800	900	75	9
900	1000	83	8
1000	1100	93	10
1100	1200	104	11

Frequency and probability of each number of arrival is shown in table-D.3 (Use MS excel formula, COUNTIF(range of cells of 4th column of table D2, '=' & cell of 1st column of table D3)) for example COUNTIF(\$K\$5:\$K\$35,"="&M6). The range of cells should be made fix by pressing F4. Here traffic intensity, $A = (1/12)*100 = \lambda T = 8.333\text{Erls}$ and theoretical probability, $P_x = (A^x/x!)e^{-A}$ is shown in table-D.4.

Index	Frequency	Probability simulation	Probability theoretical
0	0	0	0.00024
1	0	0	0.002003
2	0	0	0.008346
3	2	0.064516	0.023184
4	3	0.096774	0.0483
5	3	0.096774	0.080499
6	2	0.064516	0.111805
7	3	0.096774	0.133101
8	6	0.193548	0.138646
9	2	0.064516	0.128376
10	2	0.064516	0.10698
11	4	0.129032	0.081046




```

clear all
close all
N=2000; %Number of inter-arrival time
Lam=1/12; %Arrival Rate
ar_int=0; %initialize arrival instant
for i=1:N
    t=-(1/Lam)*log(1-rand()); % Inter-arrival time
    %Inter arrival time
    ar_int = ar_int +t;
    s(i)= ar_int; %Arrival instant
end
T=100; %Observation time
M=uint8(max(s)/T); %Number of segment of arrival instant

```

SL	CDF F(t) = rand()	Interarrival time (t)	Arrival Instant (t _i)
1	0.468521	7.585101703	7.585101703
2	0.007956	0.095852588	7.680954292
3	0.43571	6.866236391	14.54719068
4	0.095414	1.203333423	15.75052411
5	0.283396	3.998790785	19.74931489
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```

n=1;
for k=1:M % Consider 1 to Mth segments
count=0;
for i=1:N %Consider all the N=2000 arrival instant
if s(i)>=(n-1)*T && s(i)<=n*T % ith arrival instant s(i) in nth Segment
count=count+1; %Number of calls in each segment
end
end
C(k)=count; %Number of calls in kth segment
n=n+1; % Reach on next segment
end %End of M for loop

```

Time Range		Number of calls in the range C(k)
From	To	
0	100	15
100	200	3
200	300	11
300	400	11
400	500	5
500	600	9
600	700	8
700	800	4
800	900	9
900	1000	8
1000	1100	10
1100	1200	11

```

k=1;
for m=1:max(C) % For example m = 1 to 15 of the table
% 1 to maximum number of calls found considering all segments
x=0;
for j=1:M
if C(j)==m % check m number of calls in jth segment
x=x+1; % frequency of m calls
end
end
f(k)=x; % frequency of m call in the array f(k)
k=k+1;
end

```

j	Time Range		Number of calls in the range C(j)
	From	To	
1	0	100	15
2	100	200	3
3	200	300	11
4	300	400	11
5	400	500	5
6	500	600	9
7	600	700	8
8	700	800	4
9	800	900	9
10	900	1000	8
11	1000	1100	10
M = 12	1100	1200	11

%The normalized version of f gives bar graph of pdf

```
%Theory  
A=Lam*T;  
k=1:max(C);  
P=((A.^k)./factorial(k))*exp(-A);  
  
y = [f/sum(f); P];  
c = categorical({'Simulation','Theory'});  
bar(c,y)
```

